Aman Obaid

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EDUCATION

Indian Institute of Technology Madras
B.S in Data Science and Applications

SKILLS

Languages & Backend: Python, JavaScript, SQL, FastAPI, Flask, Celery, Express.js
Databases & Infrastructure: PostgreSQL, Qdrant, Redis, MongoDB, Docker, Google Cloud
AI & LLM Systems: LangGraph, LangChain, Multi-Agent Systems, RAG, ReAct Architecture

WORK HISTORY

Pinch Lifestyle Pvt Ltd Aug 2024 – Jan 2026
SDE 1

- Built an agentic AI personal assistant for **1000+ users** in **Python** and **LangGraph**, automating flight, hotel, and product research to replace a slow manual concierge workflow.
- Designed a **ReAct-style agent** in LangGraph integrating search APIs with persistent **long-term memory**, autonomously querying external vendors and synthesizing personalized options in real time.
- Built a bidirectional **AI-human handoff** system with shared conversation context, reducing repeated user inputs during handoffs by **67%** and enabling seamless transitions between AI assistance and personalized concierge support
- Built a **RAG pipeline** (**LangChain**, **Qdrant**) indexing company operating procedures to enforce pricing and brand tone, drastically reducing hallucinated recommendations.
- Shipped services with **Docker** to **Google Cloud**, instrumenting **Cloud Monitoring** and distributed tracing to identify latency bottlenecks, cutting per-turn response times by **45%** (**9.5s to 5s**) using **Redis** caching and concurrent async retrieval.

PROJECTS

AgenticBlogger – Multi-Agent Technical Content Engine GitHub

- Automated technical content publishing by architecting a **multi-agent** pipeline in **LangGraph**, **FastAPI**, and **PostgreSQL** that produces researched articles with verified code and diagrams.
- Prevented broken or hallucinated code from reaching publication by building an isolated subprocess **sandbox** with **AST** validation instead of unsafe **eval()**, feeding tracebacks to an agent to auto-patch errors.
- Cut revision token costs by **74%** and preserved approved content by rewriting only targeted sections during editorial feedback instead of regenerating the entire document.
- Eliminated workflow loss during long human-review pauses by persisting agent state in **PostgreSQL** with LangGraph, resolving cloud **PgBouncer** connection crashes via zero-statement caching.

GrabPic – AI Event Photo Discovery App GitHub

- Automated event photo discovery by building an AI platform with **FastAPI**, **PostgreSQL**, and **Qdrant** that matches attendees across bulk gallery photos from a single uploaded selfie.
- Scaled ingestion to **50 images/sec** without API lag by decoupling uploads from heavy ML inference using **Celery** and **Redis** workers for face detection and 512-dim embedding extraction.
- Delivered personalized photo results in under **200ms** with strict privacy by querying **Qdrant** via **cosine similarity** with event-level payload filtering and **JWT** authentication.